



Konatham Praveen Raj

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Electronics and Communication Engineering
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EDUCATION

- **Master of Technology, Signal Processing and Machine Learning** 2025 – Present
National Institute of Technology, Surathkal, Karnataka CGPA: 6.73 (SEM-1)
- **Bachelor of Technology, Computer Science and Engineering** 2019 – 2023
Indian Institute of Information Technology, Tiruchirappalli CGPA: 7.43

PROJECTS

- **Deep Learning for Prostate Cancer Grading (Minor Project – EC788)** 2025 – Present
M.Tech Minor Project under Prof. Shyam Lal, NITK Surathkal
 - Curated zero-leakage PANDA Challenge dataset: 302,688 patches (256×256) across 6 ISUP classes with slide-level 70/15/15 split.
 - Benchmarking CNN architectures (VGG16, ResNet50, EfficientNetB3, DenseNet121) for multiclass grading of histopathological WSI patches.
 - Stack: TensorFlow/Keras, Python, OpenSlide, RTX 4060 (8GB VRAM).
- **Automatic Modulation Classification (AMC) for Low-SNR Signals** 2025
Pattern Recognition and Machine Learning Course Project (Team)
 - Built end-to-end pipeline classifying ASK/FSK/PSK/NOISE signals under AWGN (SNR: -5 to 20 dB).
 - Engineered time-domain, frequency-domain, and phase features; trained Random Forest achieving 89.26% accuracy and 99%+ above 5 dB SNR.
 - Stack: Python, scikit-learn, NumPy, SciPy.
- **PFT-SR: Single Image Super-Resolution Implementation** 2025
Image Processing and Computer Vision Course Project
 - Reproduced Progressive Focused Transformer (PFT-SR) paper (arXiv:2503.20337) with custom CUDA op compilation and Colab-based inference pipeline.
 - Stack: PyTorch 2.5, CUDA 12.1, Python.
- **C3 Neural Compression Implementation** 2025
Information Processing and Compression Course Project
 - Reproduced Google DeepMind’s C3 learned image compression model; set up GPU-accelerated Docker environment (JAX + TensorFlow) and benchmarked on Kodak dataset.
 - Stack: JAX, TensorFlow, Docker, Python.
- **Thermal Face Detection System**
Internship Project – Cojag Smart Technology Pvt. Ltd.
 - Developed real-time multi-face detection from thermal video; implemented preprocessing, segmentation, and ML-based classification pipeline.
 - Optimized model for embedded and real-time deployment using Python and OpenCV.
- **Healthcare Security in Federated Learning (B.Tech Major Project)**
B.Tech Final Year Project, IIIT Tiruchirappalli
 - Implemented FL framework on medical datasets; analyzed gradient poisoning and model inversion attacks with corresponding defense strategies.
 - Stack: PyTorch, Python.

TECHNICAL SKILLS

Languages: C, C++, Python	Frameworks & Tools: TensorFlow, Keras, PyTorch, scikit-learn, OpenCV, JAX, Docker
OS: Linux (Ubuntu), Windows	Domains: Deep Learning, Signal Processing (FFT, filters), Image Processing, Federated Learning, FPGA (Verilog, Xilinx)

AREAS OF INTEREST

- * Deep Learning & Computer Vision
- * Signal Processing & ML
- * Medical Image Analysis
- * FPGA Design

CERTIFICATIONS

- * **Hack The League Hackathon** – Recognized for ML-based application development.
- * **Cojag Smart Technology Internship** – Thermal imaging and machine learning systems.